

1984, three years after the release of the IBM PC (the first mass-market computer). The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption thus far suggests that it has the potential to be similarly disruptive. Moreover, we show that genAI adoption by workers is proceeding at a faster pace than adoption by firms, highlighting both challenges and opportunities for managers seeking to capitalize on this new technology (Babina et al., 2024; Bonney et al., 2024; Doshi and Moore, 2025; McElheran et al., 2024).²

Second, which workers are currently using genAI, and what factors contribute to an individual’s adoption decision? We find substantial variation in genAI adoption across demographic and labor market characteristics. Adoption has been faster for male, younger, more educated, and higher-wage workers. Adoption also differs greatly by occupation, suggesting that differences in job roles and responsibilities are an important determinant of adoption. Consistent with this, we show that recent estimates of genAI task-based exposure are highly predictive of actual adoption (Eloundou et al., 2024; Felten et al., 2021), and that occupations with lower adoption also have higher shares of workers who report that genAI is not useful for their job. These findings contribute to our understanding of how genAI maps onto specific tasks and roles within the firm, which is relevant both for allocating adoption support and adapting firm organization.

We provide evidence that firm climate and policies also appear to play an important role in workers’ adoption choice. A sizable share of workers who do not use genAI cite workplace barriers as a key reason, and this share is particularly large in a few occupations who may deal with sensitive data. Alternatively, workers whose employers encourage them to use genAI are much more likely to do so, even after controlling for other characteristics. This suggests that when job tasks are amenable to genAI adoption, firms can play an active role by promoting usage, offering training, and removing organizational barriers to adoption.

Third, how large are potential productivity gains from genAI? Based on self-reported time savings due to genAI, the average genAI user would need to work 5.2% more hours in order to complete their work without genAI, implying 1.4% time savings across all workers (including non-users). We use a standard production model to estimate potential productivity gains at the industry and aggregate level. We find wide variation in potential productivity gains by industry,

²The Census BTOS Survey estimates between August and November 2024 5.4% of firms had incorporated AI technologies into their production process. This suggests that firm-based measures may miss an important share of genAI adoption. Aaron Levie, the founder and CEO of Box, a cloud storage provider, compares this pattern with past technologies: “It’s almost a one-to-one parallel with the early wave of consumerization of IT. We saw that exact same dynamic in the early 2010s, which was employees of enterprises were pretty frustrated with their traditional document management and collaboration technology, and so they would go out and just go onto a web browser, sign up for a Box account, and then bring it into their organization.” (Levie, 2024)

A key question is whether genAI’s faster adoption was driven by work or non-work use.¹⁰ The faster overall pace of genAI adoption compared with PCs is driven by non-work adoption. The middle pair of bars in Figure 2 plots work adoption of genAI and PCs. We find similar work adoption rates for genAI after two years (32%) and PCs after three years (25%). By contrast, the bottom pair of bars in Figure 2 reveals much higher non-work adoption rates for genAI (38%) compared with PCs (5%).

The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption by workers thus far suggests that this new technology could also be transformative for firms. At the same time, the choice of whether to adopt a technology depends on both its benefits and costs. PCs required purchasing expensive immobile hardware and nontrivial training. In contrast, many genAI products are free or inexpensive and user-friendly. On the one hand, this lower adoption cost likely contributes to genAI’s rapid uptake. On the other hand, this means that similar adoption rates for PCs and genAI at work do not necessarily imply equivalent benefits.

4 Which Workers Use Generative AI?

In this section, we document that early genAI adoption has not been uniform throughout the workforce. Understanding which workers are adopting genAI at higher rates provides several important insights. First, because many workers are adopting genAI ahead of firm-level adoption, variation in genAI use can help managers allocate support, training, and integration efforts. Second, we show that employer encouragement of genAI use is highly predictive of adoption, suggesting that firms and managers can influence the nature and extent of genAI use in their organization. Third, adoption patterns shed light on how genAI maps onto specific tasks and roles within the firm, which is relevant for decisions regarding job redesign and firm reorganization.

Figure 3 documents demographic heterogeneity in work adoption of genAI. The top panel shows genAI adoption is 6 percentage points higher for men. The middle panel show genAI adoption by age. Workers aged 18-49 have similar adoption rates, while adoption is lower for workers aged 50-64. The bottom panel shows that adoption rates are increasing in educational attainment. In particular, workers with a Bachelor’s degree or more use genAI at roughly twice the rate of non-college educated workers. Differences between demographic groups are primarily driven by users who use genAI every workday or at least once in the previous week.

Figure 4 plots the kernel density of genAI adoption by weekly earnings percentile. Adoption

¹⁰Unfortunately, internet adoption data do not allow us to distinguish work from non-work adoption.

rapidly (Nelson and Phelps, 1966). Third, genAI may directly increase workers’ productivity and wages. Our data do not allow us to distinguish between these explanations.

4.1 Tasks, Predicted Exposure, and Generative AI Adoption

Why do some groups of workers adopt genAI at higher rates than others? One reason could be that workers differ in their job tasks or responsibilities, and genAI is more helpful in assisting some tasks than others.

Consistent with this explanation, we first document that genAI adoption differs greatly between occupations; see Figure 5. The three occupations with the highest genAI adoption are computer and mathematical, management, and business and finance occupations (54%, 52%, and 48%, respectively) and the bottom three are blue collar, office and administration, and personal services occupations (26%, 23%, and 16%, respectively). At the same time, genAI adoption exceeds 20% in all but one occupation groups, suggesting that the technology is useful for a broad range of occupations. As with demographic differences, differences between occupations are driven by the rates of daily and weekly genAI users.

Do different adoption rates between occupations reflect differences in tasks or responsibilities? One approach to assess this is to construct task-based-exposure measures across occupations (Eloundou et al., 2024; Felten et al., 2021; Webb, 2020). For example, Eloundou et al. (2024) constructs a rubric that assigns each task in the O*NET database a zero or one corresponding to whether LLMs (a popular class of genAI) could feasibly produce large productivity gains for that task. They then construct a predicted exposure score for each occupation by aggregating task-level scores for each occupation. Using RPS data, we can ask whether these predictions are correlated with adoption. Our study is the first to evaluate these task-based “exposure” predictions using actual genAI adoption rates.¹²

Figure 6a plots an occupation group’s predicted genAI exposure from Eloundou et al. (2024) against actual genAI adoption in the 2024 RPS. We find that predicted genAI exposure is highly correlated with actual genAI adoption ($\rho = 0.65$). In Figure 6b, we find a similar correlation ($\rho = 0.71$) using the predicted exposure score from Felten et al. (2021). For example, personal service and blue-collar occupations have relatively low predicted exposure and also relatively low adoption, while computer and mathematical occupations have both high predicted exposure and high adoption.

The high correlation between predicted exposure and actual adoption provides partial validation of task-based exposure predictions, supporting their utility for researchers. At the same

¹²Handa et al. (2025) assign tasks to conversations by Claude.ai Free and Pro users and show that task usage rates are correlated with the predictions by Eloundou et al. (2024).